

MLP-LAB · MODEL DEEP-DIVE

Gemma 4 12B Unified

The Encoder-Free Vision-Language Model

How a 35M linear projection replaces a 150M Vision Encoder —
and how one decoder learns to see, hear, and read.



Recap: the standard VLM has a separate ‘eye’

Every model we saw — LLaVA, BLIP-2, Gemma 3 — bolts a heavy vision encoder in front of the LLM.

The two-stage pipeline



THE HIDDEN COST

The encoder is a second model.

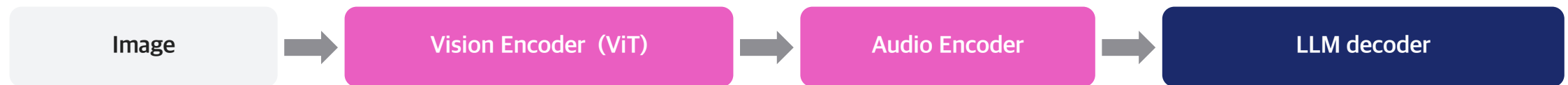
It must be loaded, stored, and co-tuned separately. **It also compresses the image before the prompt is even seen** — so prompt-relevant detail can be lost.

Question for today: can we just delete it?

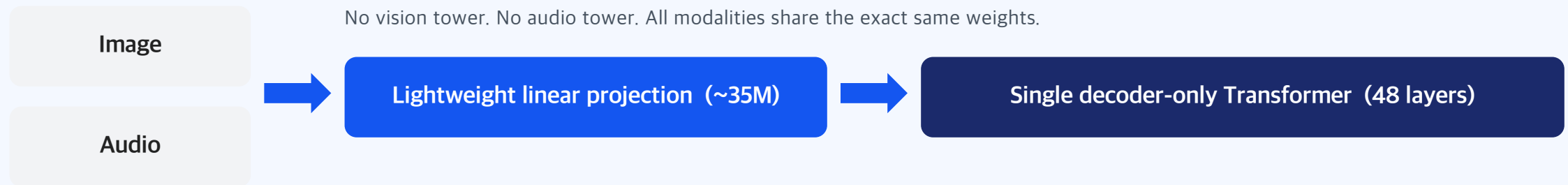
Gemma 4 12B “Unified”: delete the encoder

Raw patches and waveforms are projected straight into the decoder — one model, all modalities.

Other Gemma 4 (E4B / 31B)

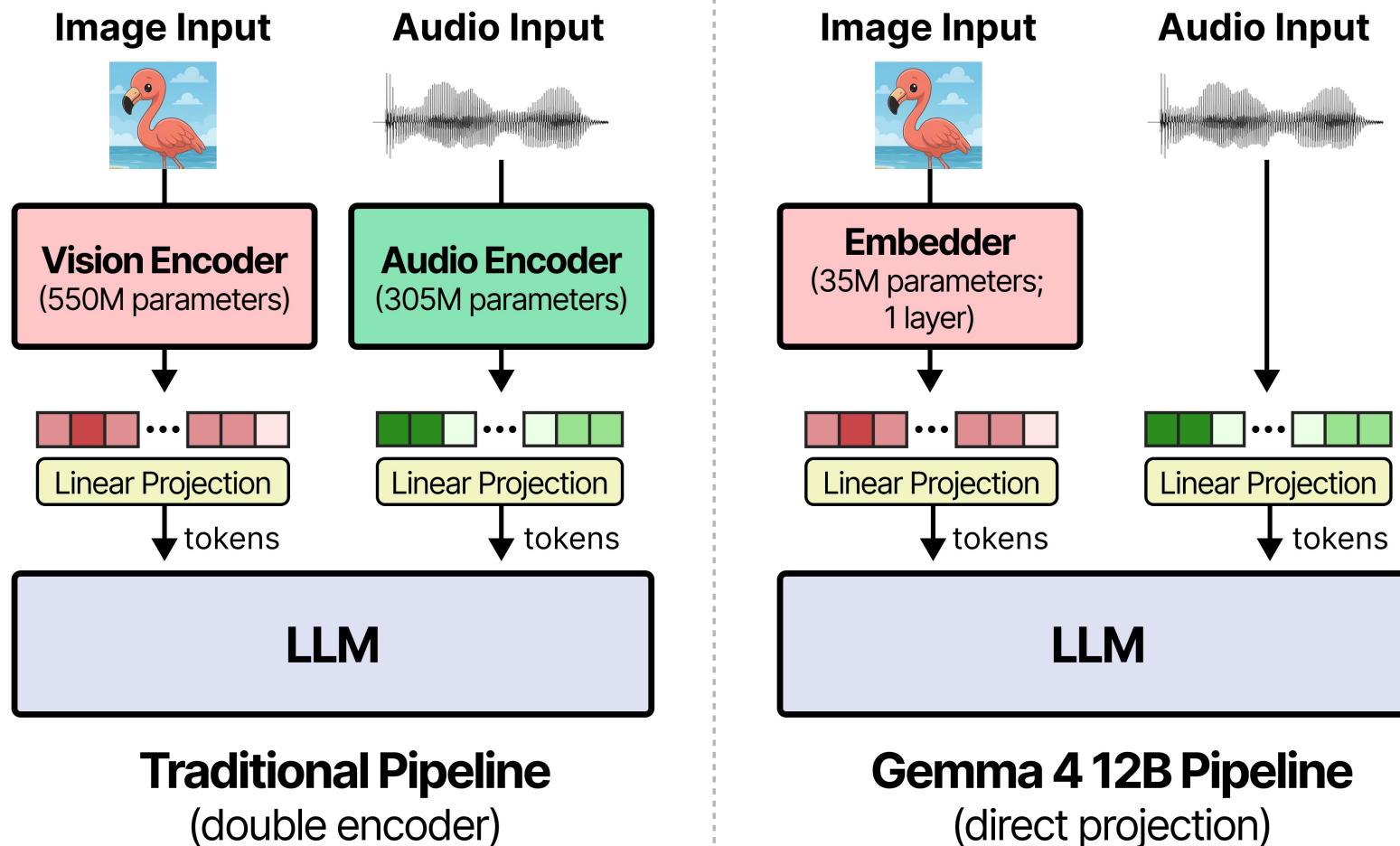


Gemma 4 12B Unified



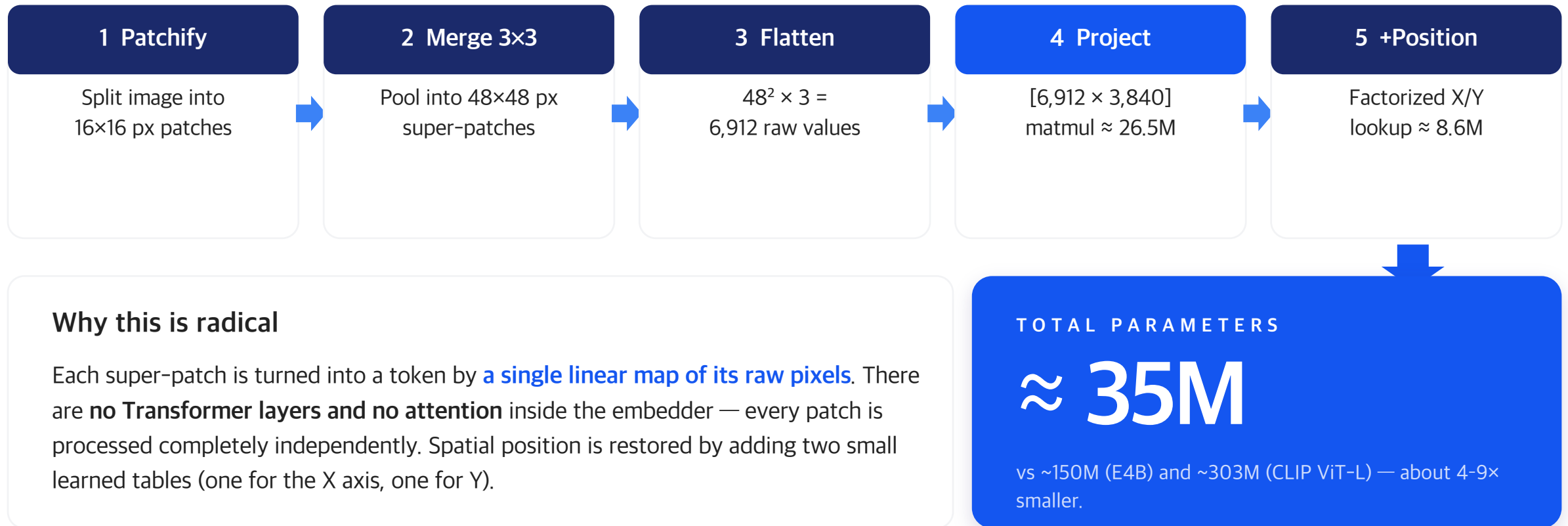
Gemma 4 12B “Unified”: delete the encoder

Raw patches and waveforms are projected straight into the decoder — one model, all modalities.



Inside the vision embedder: pixels → tokens

No attention, no layers — just one matrix multiply per patch. Verified from the vision config.



What a linear map loses — and what restores it

A single projection cannot mix patches. So the work moves into the decoder body itself.

What the encoder used to do

Cross-patch attention

16+ layers let “the patch on the right is an eye, below is a mouth” emerge.

Hierarchical features

Edges → textures → parts → objects, built layer by layer.

A linear map is one step — no nonlinearity, patches stay independent.

The fix: bidirectional attention

A decoder normally uses a **causal mask** (each token sees only the past). That is wrong for an image — the top-left patch could never see the rest.

Gemma 4 sets `use_bidirectional_attention = “vision”`: image tokens attend to **all other image tokens**, while text stays causal. The cross-patch mixing that lived in 16 encoder layers is now **absorbed by the decoder’s 48 layers**.

text → causal · image ↔ bidirectional

Audio is encoder-free too

No mel-spectrogram, no Conformer — raw 16 kHz waveform is sliced and projected directly.

From waveform to tokens

- 1 Raw 16 kHz waveform**
amplitude values, no spectrogram
- 2 Slice into 40 ms frames**
640 samples per frame
- 3 RMSNorm → Linear**
one frame → one soft token

No downsampling → #tokens = $\text{ceil}(\text{samples} / 640)$.

WHAT WAS REMOVED

12 Conformer layers ~300M params

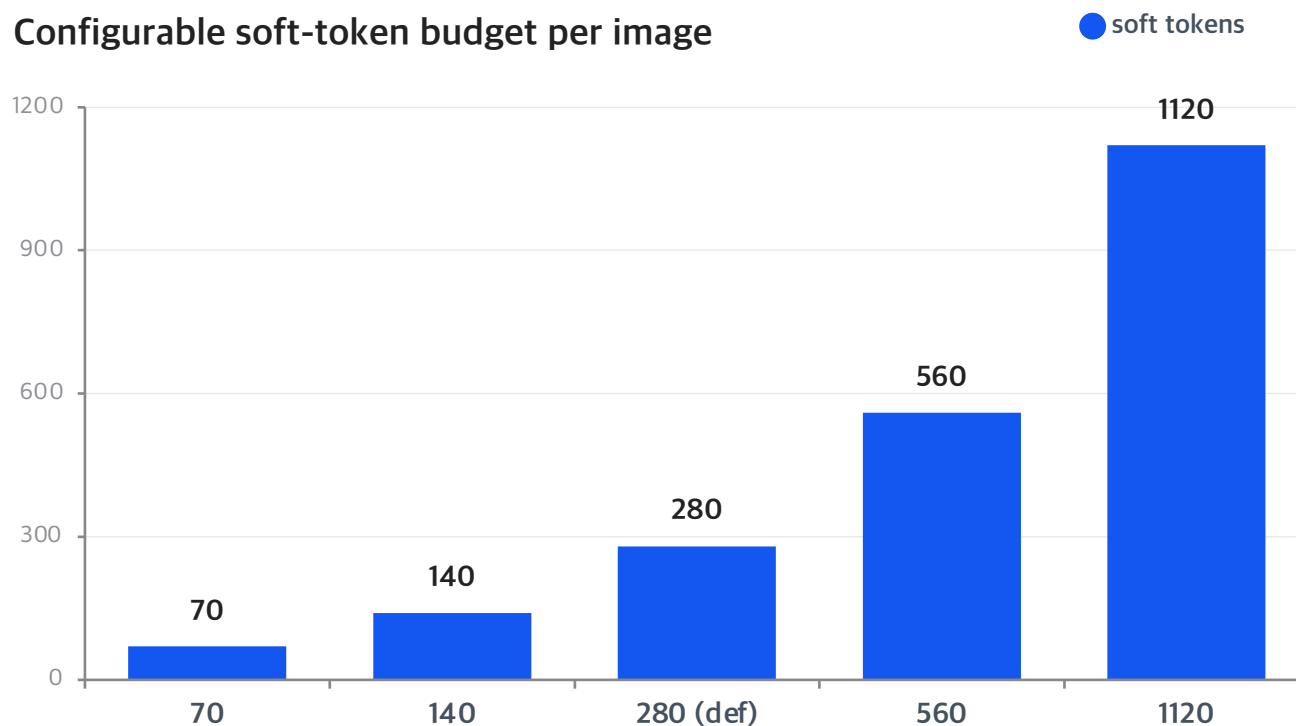
The same Gemma4UnifiedMultimodalEmbedder (RMSNorm → Linear) handles both vision and audio.

Position is free: audio is 1-D in time, so the decoder's own RoPE supplies order — no X/Y tables needed.

Encoder-free is not free: the token budget

Removing the encoder doesn't make images cheap — KV-cache and compute still scale with tokens.

Configurable soft-token budget per image



How to choose

Low budget (70-140)

classification, captioning, video frames — speed first.

High budget (560-1120)

OCR, document parsing, small text — detail first.

Key idea variable resolution is handled by token count, not by a fixed-size encoder input.

This idea has a lineage

Encoder-free VLMs go back to 2023. Each hit a wall; Gemma 4 borrows the simplest recipe.

Fuyu-8B

Adept · 2023

First practical encoder-free VLM. Linear-projected patches, but causal-only attention → weak spatial reasoning (MMBench 10.7%).

EVE

NeurIPS 2024

Showed training collapses without staged learning: freeze the LLM first, align the projection, then unfreeze.

EVEv2 / Mono-InternVL

2025

Per-modality Q/K/V or vision FFN experts finally matched encoder models — at the cost of complexity & huge data.

Gemma 4 12B

Google · 2026

Keeps Fuyu's simplicity, adds bidirectional attention + factorized position. No per-modality split.

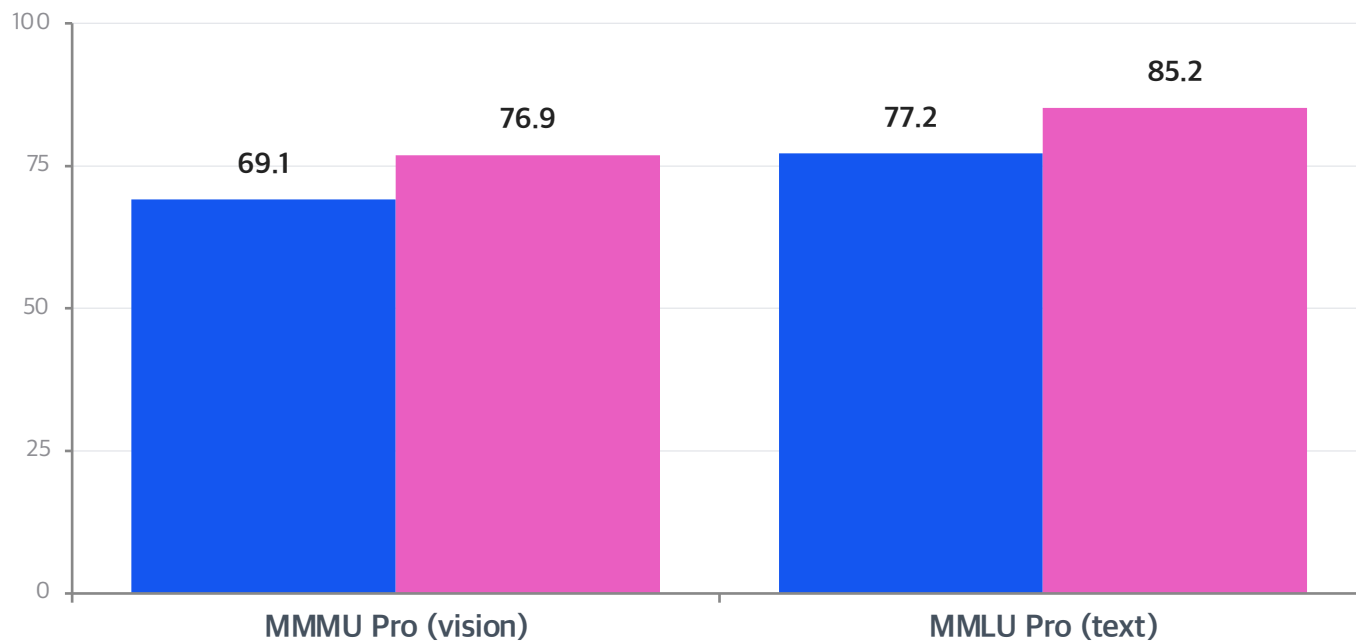
Does it work? The honest picture

Strong vision scores, but the gap to encoder models tracks model size — not the missing encoder.

12B Unified vs 31B Dense (encoder-equipped)

● 12B Unified

● 31B Dense



Read it carefully

The vision gap (≈ 7.8 pts) is almost identical to the text gap (≈ 8 pts).

Since text needs no encoder, the difference is mostly **raw model scale (12B vs 31B)** — not a penalty for being encoder-free.

Takeaway at 12B scale, deleting the encoder costs little while removing $\sim 450\text{M}$ params.

SUMMARY

One decoder to see, hear, and read

1 Delete the encoder

Vision & audio towers (~450M) replaced by a ~35M linear projection.

2 Move the work inside

Cross-patch reasoning is absorbed by bidirectional attention in the 48-layer decoder.

3 Scale pays the bill

A bigger LLM body + factorized position embeddings recover most of the lost accuracy.

4 One-pass fine-tuning

All modalities share weights — LoRA or full tuning updates the whole loop at once.

Caveat: Google has not released a Gemma 4 Technical Report — the exact training recipe (pre-alignment, encoder distillation) is unconfirmed.

Thank you

